

A Statistical Analysis of Wildland and Prescribed Fires in the Changing California Climate

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Motivation

- Climate change-driven large, destructive wildfires becoming more frequent, especially in California [1,2]

- Prescribed burns are known to be effective in reducing wildfire intensity in some scenarios [3], yet:
  - Environmental trends indicate the window for prescribed burns is shrinking
  - Limited implementation due to controversy on risk, cost, and environmental impact

- Much of the existing body of research was performed in the previous century and did not account for climate change's impact on prescribed burn efficacy

Statistical analysis can be used to identify effective or ineffective prevention strategies, such as prescribed burns, but the data needed to study this question is **fragmented** and **complex**

Research Questions

1. Are prescribed burns effective at reducing the size of wildfires?

2. Are prescribed burns effective at reducing the size of wildfires while controlling for climate-relevant variables

3. How do these relationships change at different spatio-temporal scales?

We Investigate these questions with:

- Monthly Time Series Analysis – Correlation and Cross-Correlation
- Inferential Statistics - Mann-Whitney U-test
- Regression Analysis – Ordinary Least Squares Linear Regression

Data

Variable	Source	Description	Range	Units	Symbol
Acres Burned	CALfire[4]	Wildfire acres burned	0+	Log(Acres)	log_acres
Treatment Acres	CALfire[5]	Prescribed fire acres burned	0+	Acres	rx_acres
Precipitation	NOAA[6]	Monthly Precipitation averaged across the county	0+	Inches	pcp
Temperature	NOAA[6]	Average, minimum, and maximum temperature averaged across the county	0+	Fahrenheit	temp
Palmer Index	NOAA[6]	Drought severity indices	[-10, 10]	N/A	zndx, pdsi, phdi, pmdi
Windspeed	OpenMeteo [7]	Maximum windspeed on fire start date	0+	Km/h	max_wind

Table 1: Variables in the dataset, including descriptions, sources, and units

Dataset Creation

- Created by combining NOAA climate data and CALFire fire history data
- Data from 2041 wildfires spanning January 2013 to December 2022
- Columns:
  - wildfire acres burned, precipitation, temperature, wind speed, drought index
  - for each 3-36 month time frame, get prescribed fires in 5 - 100 km distance radius

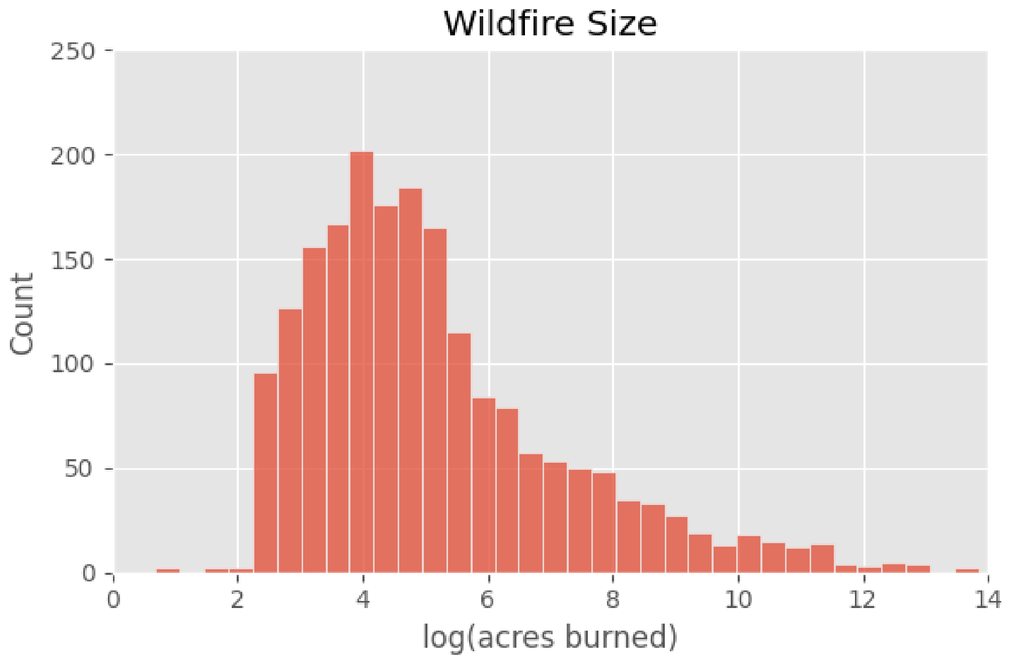


Figure 1: Distribution of wildfire size

Methods

- **Time series analysis:**
  - Correlation to identify which predictors are associated with acres burned
  - Cross-correlation to identify at what time lags predictors are associated with acres burned
- **Inferential analysis:**
  - Difference of median test to examine if wildfires are smaller when there is a prescribed fire nearby
- **Regression Analysis:**
  - Predict acres burned per wildfire incident from prescribed fires and climate variables

Time Series Analysis: Monthly

Wildfire acres burned have the strongest correlation with climate variables

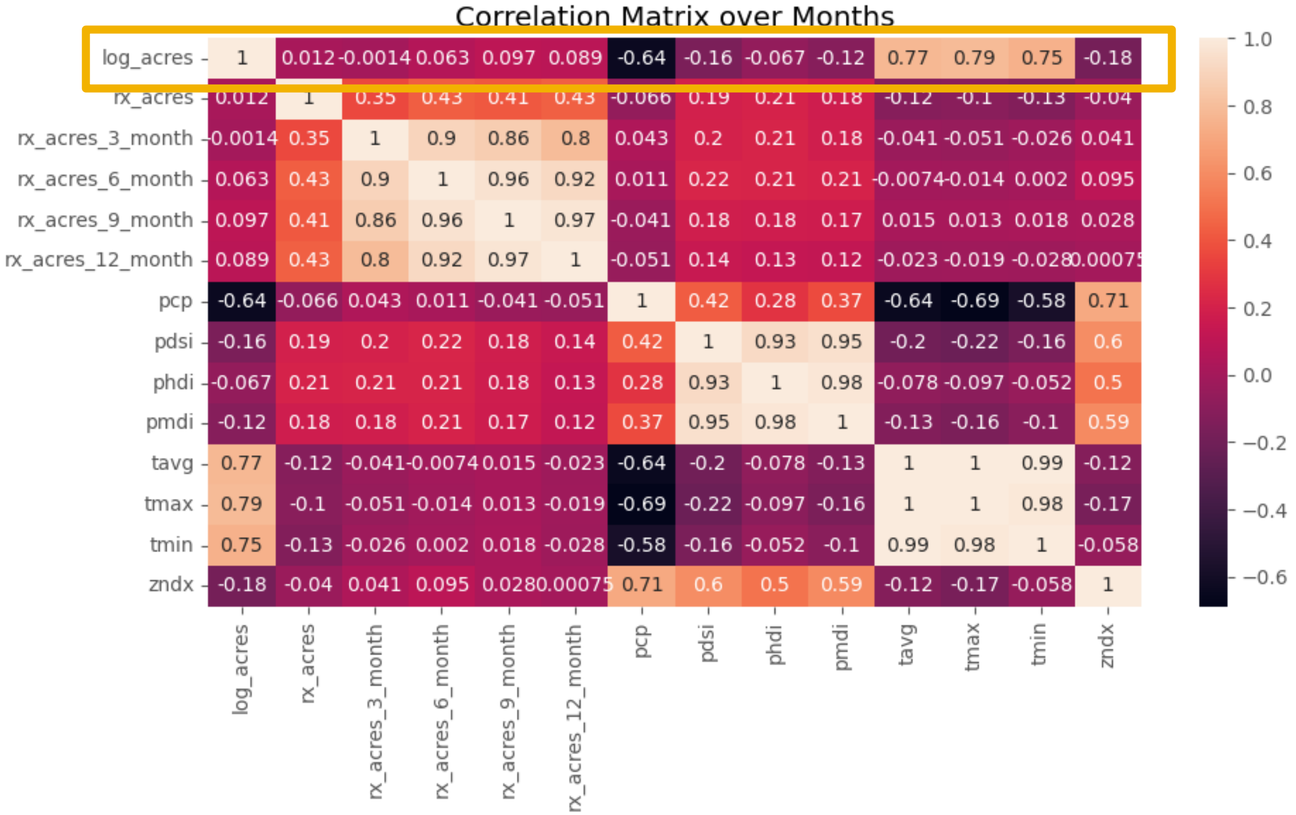


Figure 2: Correlation matrix for predictors and wildfire size

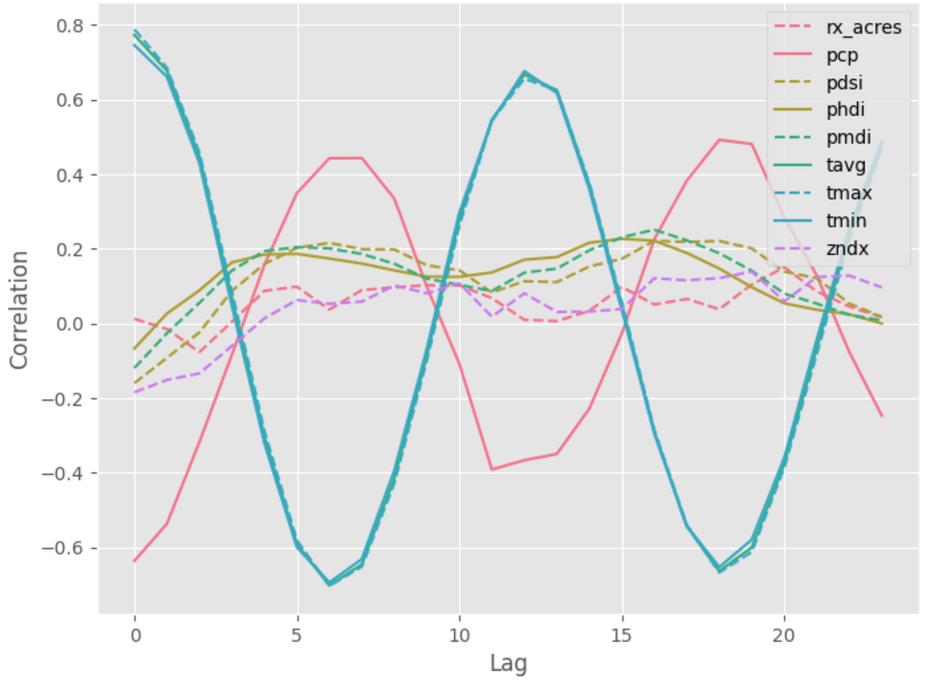


Figure 3: Cross-correlations for predictors and wildfire size

Inferential Analysis: Treatment Data

Figure 4: Box plot of wildfire size between groups with and without prescribed burns within the specified distance

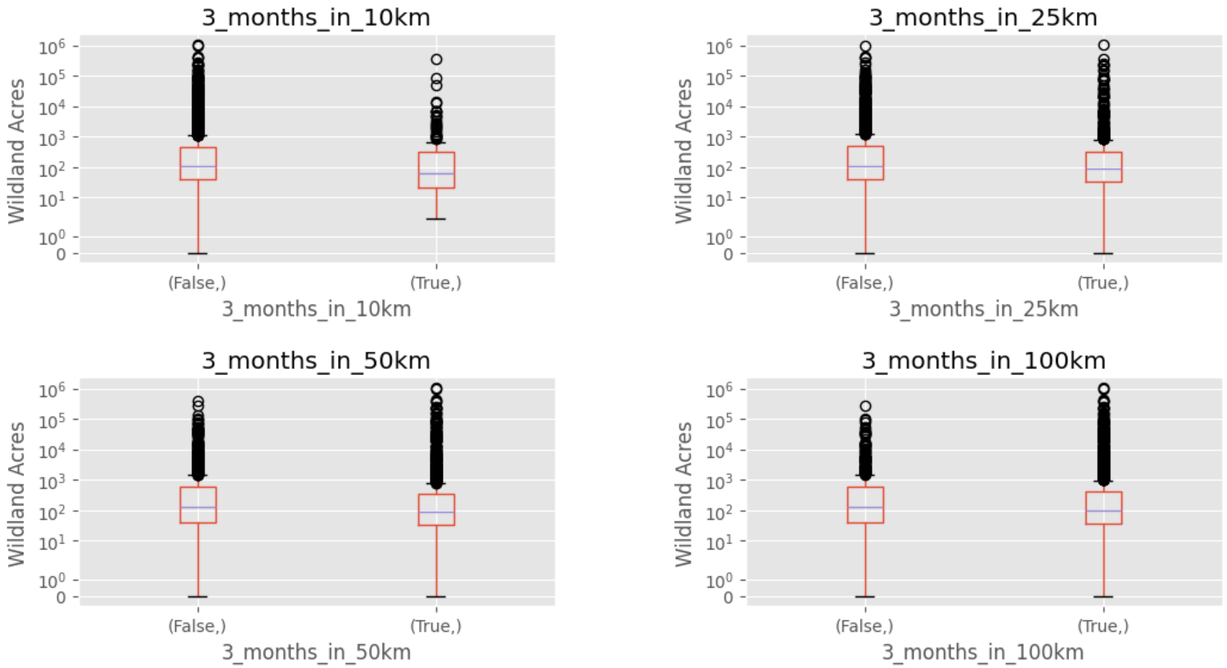


Table 2: Mann-Whitney U-test results, including rank biserial correlation (RBC) and common language effect size (CLES)

		U-val	alternative	p-val	RBC	CLES
prescribed_3_months_in_10km	MWU	116462.5	less	0.002	0.142	0.571
prescribed_3_months_in_25km	MWU	398860.5	less	0.015	0.062	0.531
prescribed_3_months_in_50km	MWU	470156.5	less	0.000	0.085	0.542
prescribed_3_months_in_100km	MWU	324331.0	less	0.044	0.054	0.527

Prescribed burn treatment is associated with fewer acres burned at 10 km, 25 km, 50 km, and 100km distance when completed 3 months before a fire

Regression Analysis: Treatment Data

Model: $\hat{Y} = \hat{\alpha} + \hat{\beta}\boldsymbol{X}$

- $R^2 = 0.885$
- $Train\ MSE = 3.476$
- $Test\ MSE = 4.535$
- $F = 747.8, \mathrm{p} < 0.001$

Prescribed burns are related to decreases in wildfire size

Decreased precipitation is related to increases in wildfire size

Region has an effect on wildfire size

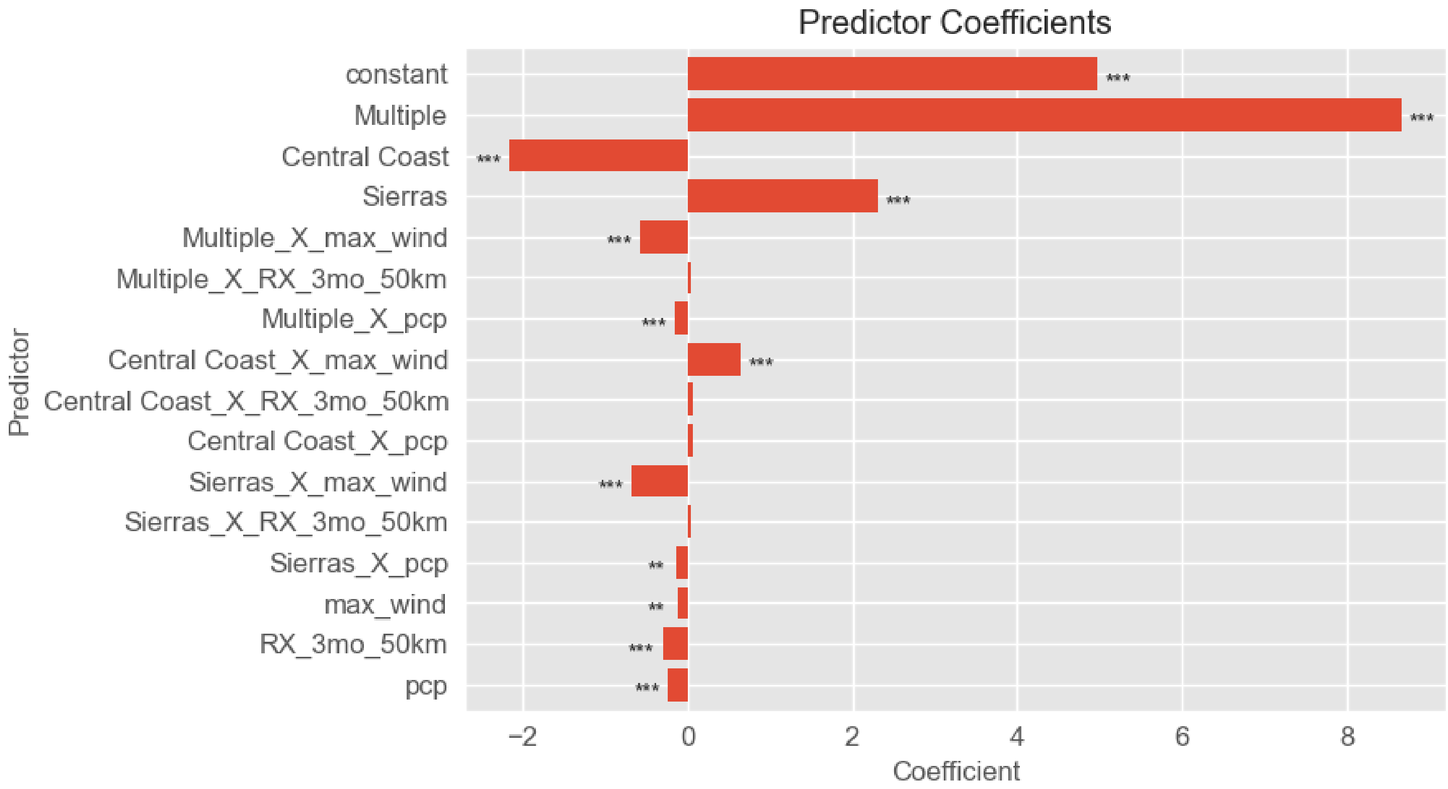


Figure 5: Regression coefficient results, with significant predictors notated with "*"

Recall: Research Questions

1. Are prescribed burns effective at reducing the size of wildfires?

2. Are prescribed burns effective at reducing the size of wildfires while controlling for climate-relevant variables

3. How do these relationships change at different spatio-temporal scales?

Results indicate:
- Increases in prescribed burn acreage are related to decreases in wildfire size
- The effect of prescribed burns is significant when controlling for precipitation
- The time and location of prescribed burns impacts efficacy

Discussion and Conclusions

- Increased prescribed burns are associated with decreased wildfire size for certain cases
- When controlling for climate variables, prescribed burns are still a significant predictor of wildfire size
- Prescribed burns have the greatest effect when performed within 3 months and 50 km of wildfires when controlling for precipitation and wind

Limitations:

- Vegetation type is not considered in this model
- Models use binning for handing spatial and temporal differences – an integrated model could be better

References

[1] CALFire. Top 20 Most Destructive California Wildfires. Accessed on 2023-01-05. url: https://www.fire.ca.gov/our-impact/statistics

[2] CALFire. Top 20 Largest California Wildfires. Accessed on 2023-01-05. url: https://www.fire.ca.gov/our-impact/statistics.

[3] Molly E Hunter and Marcos D Robles. “Tamm review: The effects of prescribed fire on wildfire regimes and impacts: A framework for comparison”. In: Forest Ecology and Management 475 (2020), p. 118435

[4] CALFire. Incident Data. 2023. url: https://www.fire.ca.gov/incidents/

[5] CALFIRE. Prescribed Burns. 2022. url: https://data.ca.gov/dataset/prescribed-burns1

[6] NOAA National Centers for Environmental information. Climate at a Glance:Statewide Time Series. 2023. url: https://www.ncei.noaa.gov/access/monitoring/climate-at-a-glance/statewide/time-series

[7] Zippenfenig, P. Open Meteo Historic Weather API. 2024. doi: https : / / zenodo.org/records/7973572. url: https://open-meteo.com/en/docs/historical-weather-api



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Regression Analysis: Treatment

Figure 6: Residuals and partial regression for precipitation

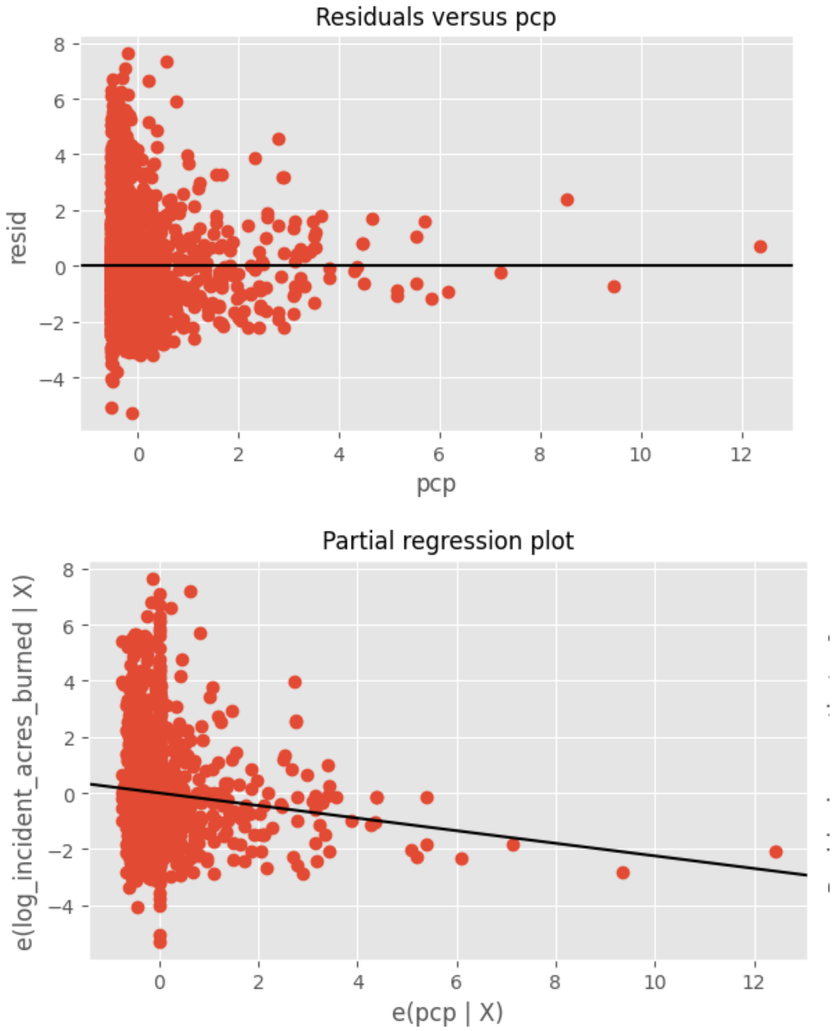
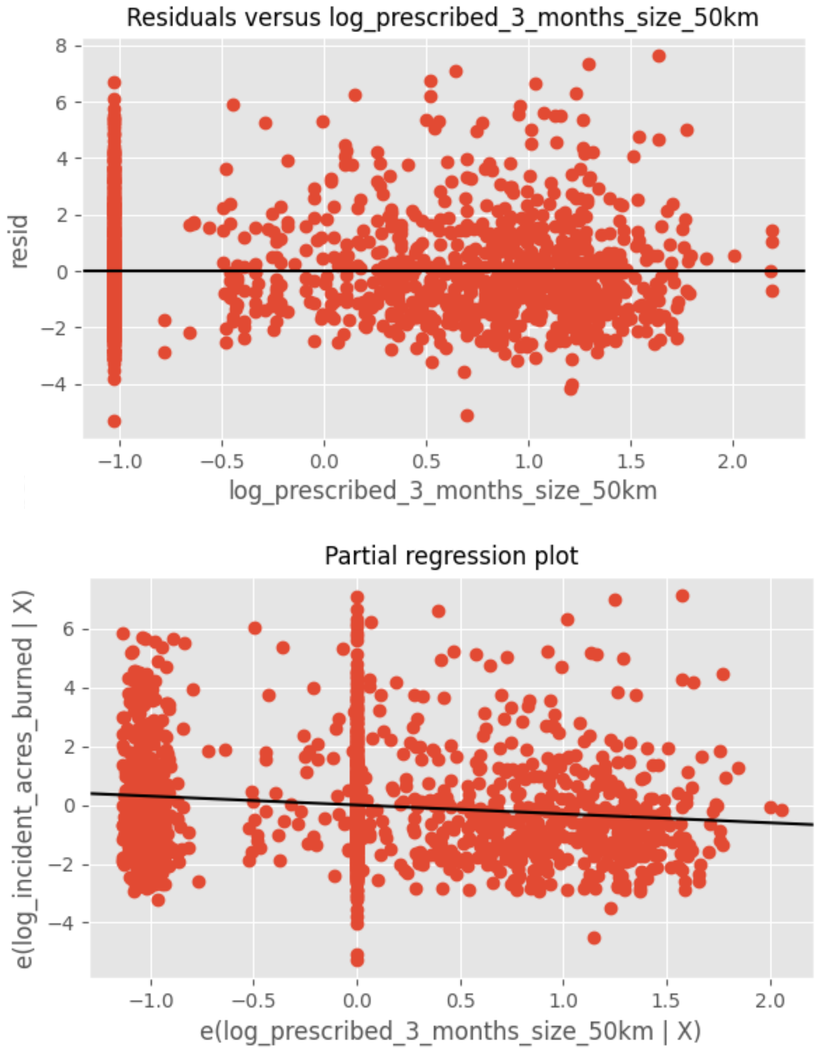


Figure 6: Residuals and partial regression for prescribed burns



Methods

- **Time series analysis:**
  - Correlation to identify which predictors are associated with acres burned
  - Cross-correlation to identify at what time lags predictors are associated with acres burned
- **Inferential analysis:**
  - Difference of median test to examine if wildfires are smaller when there is a prescribed fire nearby
- **Regression Analysis:**
  - Predict acres burned per wildfire incident from prescribed fires and climate variables

Inferential Analysis: Treatment Data

Figure 5: Box plot of wildfire size between groups with and without prescribed burns within the specified distance

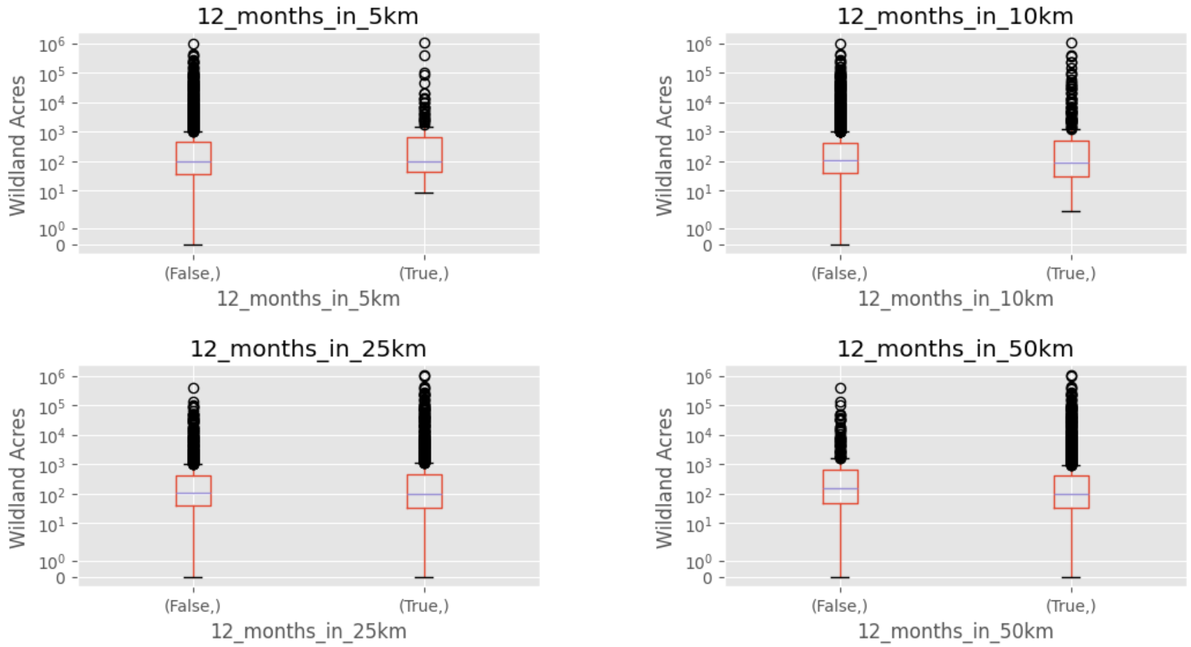


Table 3: Mann-Whitney U-test results, including rank biserial correlation (RBC) and common language effect size (CLES)

		U-val	alternative	p-val	RBC	CLES
prescribed_12_months_in_5km	MWU	136871.0	less	0.618	-0.015	0.492
prescribed_12_months_in_10km	MWU	298599.0	less	0.039	0.058	0.529
prescribed_12_months_in_25km	MWU	500004.5	less	0.194	0.022	0.511
prescribed_12_months_in_50km	MWU	214617.0	less	0.000	0.132	0.566

Prescribed burn treatment is associated with fewer acres burned at 10 km 50 km distances when completed 12 months before a fire

Data Overview

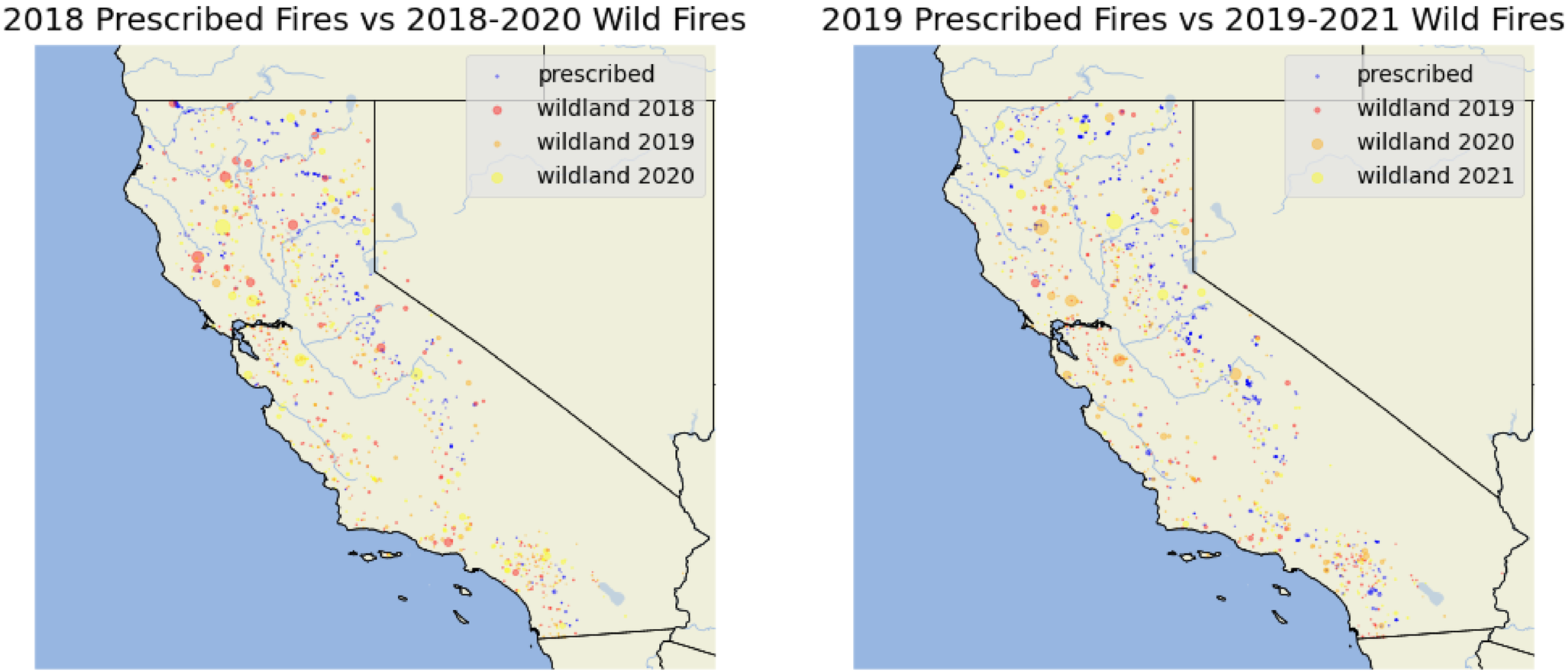


Figure 2: Visualization of prescribed fires for each year and wildfires in following years

Regression Analysis: Drought Index

$$Y = -0.615 + 1.392x_t + 2.543X_p + 0.327X_s$$

Y: PZI drought index

X: maximum temperature, precipitation, soil moisture remove correlated variables

Temperature increases are related to increased drought severity

Precipitation and soil moisture decreases are related to increased drought severity

Precipitation has the greatest impact on drought index

- $R^2 = 0.753$
- Train $MSE = 1.303$
- $F = 81.49, \mathrm{p} < 0.001$

	coef	std err	t	P>|t|	[0.025	0.975]
temp_max_Value	1.3915	0.181	7.687	0.000	1.031	1.752
precip_new_Value	2.5432	0.186	13.646	0.000	2.172	2.914
soil moisture	0.3268	0.140	2.331	0.022	0.048	0.606
intercept	-0.6151	0.125	-4.939	0.000	-0.863	-0.367

Table 5: Regression coefficient results

Regression Analysis: Drought Index

Figure 9: Residuals and partial regression for temperature

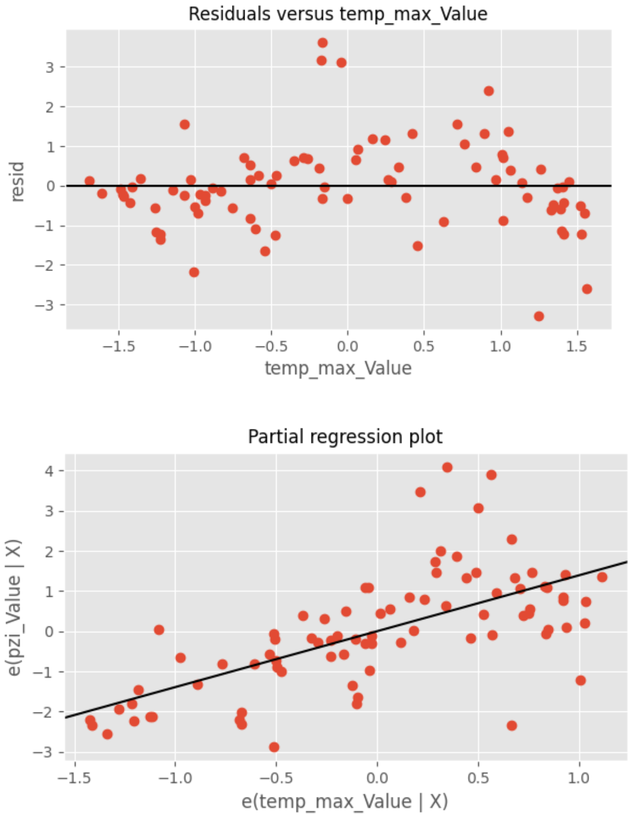


Figure 10: Residuals and partial regression for precipitation

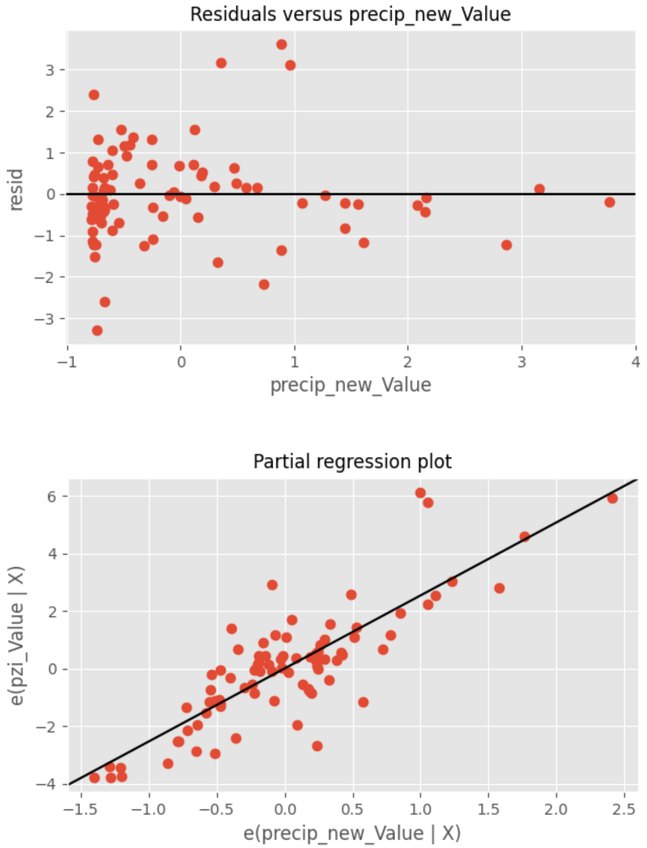
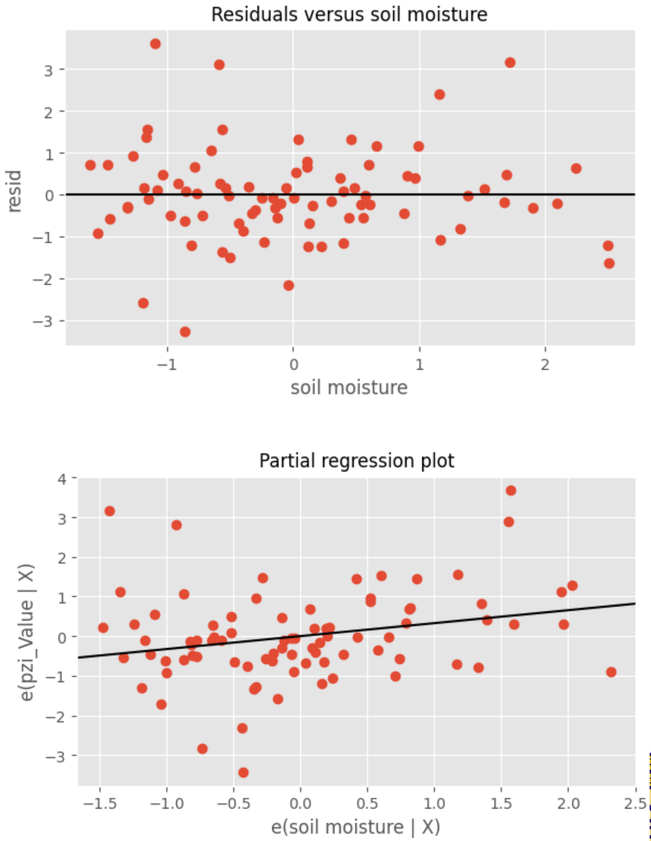


Figure 11: Residuals and partial regression for soil moisture



Discussion

- Prescribed burns are associated with decreased wildfire size
- Severe drought is associated with larger wildfire size
- Latitude and longitude are significant predictors, indicating spatial location is important

Limitations:

- Vegetation type is not considered in this model
- Models use binning for handing spatial and temporal differences – an integrated model could be better